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Novel Application of Multifrequency Dielectric Technique for Hydraulic Fracture Characterization in Different Rock Formations

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Abstract

The multifrequency dielectric technique has emerged as a promising petrophysical tool for estimating water saturation and calibrating Archie parameters, with growing potential for fracture characterization. However, most previous studies in this area have focused on numerical models or synthetic systems. This study presents a novel experimental approach for characterizing hydraulic fractures in sandstone and carbonate formations using multifrequency dielectric measurements under both dry and brine-saturated conditions. Rock samples were subjected to routine core analysis to determine porosity, permeability, and mineralogy. Nuclear Magnetic Resonance (NMR) was employed to assess pore systems and porosity distribution. An open-ended coaxial dielectric probe, operating between 1 MHz and 3 GHz, was used to measure relative permittivity and electrical conductivity before and after inducing hydraulic fractures with controlled vertical and horizontal orientations. Measurements were taken at various distances from the fracture (near and far) and under dry and brine-saturated states using 3 wt.% KCl brine. The results demonstrate that dielectric dispersion is highly sensitive to fracture presence and geometry. The permittivity was increased by 22 units on average, from around 45 to more than 60 for carbonate rocks, while the increase varied between 15 and 35 units for sandstone rocks. Also, measurements taken near the fracture showed stronger dielectric responses due to enhanced fluid connectivity and polarization effects. The conductivity increased by around 12 to 34% in fractured carbonate and sandstone samples, respectively. Fracture orientation had a notable impact, with vertical fractures producing relatively higher conductivity and permittivity compared to horizontal fractures. In contrast, NMR was unable to distinguish fracture orientation due to its bulk measurement nature. This

study establishes a robust and experimentally validated method for characterizing hydraulic fractures using multifrequency dielectric measurements. The approach offers enhanced resolution of fracture orientation, connectivity, and localization, with significant implications for both conventional and unconventional reservoir development.

Keywords: Dielectric dispersion, fracture orientation, carbonate, sandstone, permittivity, conductivity

Introduction

Dielectric measurements were introduced into the petroleum industry around the 1970s for determining the water saturation (Geng et al., 1983; Hizem et al., 2008; Al-Heeti and Al-Fatlawi, 2022). It is particularly effective in distinguishing between water and hydrocarbon-bearing zones due to the significant contrast in their dielectric constants, for a wide range of water salinity (Herlinger, 2019). The dielectric properties are associated with the charge alignment produced by the excited electric field. Consequently, the polarity of water molecules allows for the estimation of the volume of water present in the reservoir (Taherian et al., 1990; Herlinger, 2019). Dielectric tools have undergone substantial development over the years, and they are used in many industries nowadays. Though considerable work has been done in the field (e.g., Worthington and Collar, 1984; Ilesmes and Morgan, 2001; Tabbagh et al., 2009; Zisser and Nover, 2009; Weller et al., 2010; Seleznev et al., 2011; Peksa et al., 2017; Han et al., 2019; Han and Yang, 2019; AlZoukani et al., 2024), there are still some gaps that need to be addressed. For example, there are no established studies that relate the frequency-dependent dielectric response of fracture sedimentary rocks to the connectivity of fractures, types of fluids within the fractures, and the dielectric anisotropy (Han and Yang, 2019). Furthermore, there is currently no wettability index based on dielectric measurements (Oshaish et al., 2023). The goal of this work is to expand and work further in developing dielectric tools that contribute directly to the oil and gas industry. To elaborate further, this study investigates the effects of hydraulic fracturing (HF) on the dielectric properties of carbonate and sandstone.

Permittivity gauges the degree of a medium's responsiveness to an electric field influence. Three primary physical phenomena contribute to permittivity: the displacement of the electronic cloud around atoms, the coherent alignment of pre-existing microscopic dipoles, and the polarization effects at interfaces (Hizem et al., 2008). Permittivity (ϵ), also known as the dielectric constant, is a macroscopic property of a medium that relates to the electric flux density generated by an induced electric field (Kremer and Schönhal, 2002; Herlinger, 2019). Permittivity directly reflects a medium's sensitivity to the influence of an electric field. Therefore, determining the permittivity can provide an important insight into the dielectric material. Courtney (1970) stated that for microwave frequencies (> 300 MHz), different techniques are used to determine the complex permittivity. Examples include methods that depend on the standing wave within the dielectric, approaches that rely on the reflected waves from the dielectric, and techniques that depend on transmitted waves. These distinct types of measurement are also applied to different scenarios, depending on the given state material. To illustrate, the first type is suitable for moisture dielectric, while the third type is suitable for materials with a large dielectric constant. Al-Harashseh et al. (2009) stated that dielectric measurement can be conducted through many methods, depending on factors that include the material and the state of the material, whether it is saturated or dry. Also, they showed that the dielectric properties of an oil shale were obtained through the cavity perturbation technique, where at a specific frequency electric field was applied to the oil shale to calculate the dielectric constant and the loss factor. Moreover, the coaxial probe technique was used to obtain the dielectric characteristics of the oil shale at varying frequencies (Al-Harashseh et al., 2009), which is very useful for reservoir characterization. Furthermore, Alarifi and Mahmoud (2022) mentioned that dielectric properties are key metrics that contribute to evaluating existing reservoirs or help to discover a new reservoir of oil and gas. Also, they reported that dielectric properties are significant contributors not only in finding new reservoirs but also in enhancing the performance of existing reservoirs as well as optimizing the well design. This indicates that the process of evaluating and

enhancing established reservoirs can be more efficient, as the dielectric response of samples from the well can be used to characterize the reservoir.

Multifrequency dielectric measurements offer a powerful, non-destructive method for characterizing hydraulic fractures in subsurface reservoirs (Sentenac et al., 2018). By probing the dielectric permittivity and electrical conductivity of rocks over a range of frequencies, this technique enables the detection and evaluation of fracture presence, orientation, and fluid content (Shen et al., 1985; Oshaish et al., 2023). Hydraulic fractures alter the dielectric properties of rocks, particularly when filled with conductive or polar fluids such as brine or hydrocarbons (Beloborodov et al., 2017). At lower frequencies, dielectric responses are influenced by interfacial polarization and bulk conductivity, which are sensitive to fluid-filled fracture networks and matrix conductivity. Higher frequencies, on the other hand, provide insight into the rock matrix and pore-scale features by minimizing polarization effects and capturing more intrinsic material properties (Chelidze et al., 1999; Han and Yang, 2018; Han et al., 2019).

The frequency-dependent behavior (dielectric dispersion) helps distinguish between different fracture conditions: open versus mineralized fractures, and dry versus fluid-filled fractures. This multifrequency approach also enhances spatial resolution and sensitivity to fracture connectivity and anisotropy, making it especially useful for evaluating stimulation effectiveness in unconventional reservoirs. Similarly, nuclear magnetic resonance (NMR) measurements at different frequencies can help differentiate between mobile and bound fluids in fractured zones (Golsanami et al., 2016; Elsayed et al., 2022). In summary, multifrequency dielectric techniques provide a detailed, scale-sensitive characterization of hydraulic fractures, supporting better reservoir modeling, fluid flow prediction, and enhanced hydrocarbon recovery strategies.

In this study, the impact of hydraulic fracturing on the dielectric properties was studied for sandstone and carbonate rock samples. Controlled vertical and horizontal fractures were induced in the samples, and their dielectric responses were measured before and after applying the hydraulic fracturing. The experiments were conducted in two phases: initially, dielectric measurements were performed on dry rock samples; subsequently, the samples were saturated with KCl brine and re-measured to evaluate the influence of fluid content on their dielectric response. The primary objective of this study is to assess the effects of hydraulic fractures on the dielectric behavior of sandstone and carbonate reservoir rocks under both dry and saturated conditions. To deepen the understanding, multifrequency dielectric results were integrated with nuclear magnetic resonance profiles to provide more insights into the effects of fracture orientation and spacing on the dielectric and NMR profiles. This understanding will aid in the development of more accurate models for predicting the dielectric properties of fractured formations, ultimately contributing to improved hydrocarbon reservoir evaluation and management.

Materials and Methods

In this work, carbonate and sandstone samples were subjected to a systematic workflow as part of the experimental phase. The process began with hydraulic fracturing (HF), where controlled fractures were introduced in various orientations. Following fracturing, dielectric measurements were conducted on the dry samples using the open-ended coaxial probe technique. Thereafter, the samples were saturated with 3 wt.% KCl brine under a confining pressure of 1900 psi for 24 hours, to ensure full fluid saturation. After saturation, dielectric measurements were repeated under the same technique and conditions. All measurements were conducted at ambient conditions and repeated multiple times to ensure the consistency and reliability of the results. Finally, the collected data were processed and analyzed to evaluate the effects of hydraulic fractures and fluid saturation on the dielectric behavior of the samples. The following sections provide a detailed description of the materials used and the experimental procedures carried out in this study.

Materials

Two types of rocks were used in this study: Indiana limestone (IND) and Bandera sandstone (BND). Dielectric measurements were conducted on both rock types before and after hydraulic fracturing, under dry and saturated conditions. To simulate different fracture orientations and distances, two sets of samples were prepared: in the first set, a vertical fracture was induced at the midpoint of the sample, while in the second set, a horizontal fracture was introduced one-third of the way from the inlet face. For the saturation phase, a 3 wt.% KCl brine solution was employed, and the samples were subjected to a confining pressure of 1900 psi for 24 hours to ensure full saturation. Baseline dielectric measurements were acquired prior to saturation and fracture induction, and dielectric permittivity and electrical conductivity profiles were recorded. Additionally, for the saturated samples, nuclear magnetic resonance (NMR) measurements were performed to provide complementary insights into fluid distribution and pore structure. Moreover, the rock mineralogy was determined for the samples using XRD analysis. Indiana limestones were primarily composed of 97% calcite (CaCO_3) with minor amounts of clay minerals, quartz, dolomite, and low organic content. It typically features a fine to medium-grained, fossiliferous texture with an average porosity of 18% and permeability of 12 mD. On the other hand, Bandera sandstones consisted mostly of quartz (70%) with cementing materials such as clay minerals. The sandstone was usually fine to medium-grained, well-sorted, and had an average porosity of 19.5% and permeability of 27 mD. Table 1 provides an overview of rock samples and hydraulic fracture orientation applied in this study.

Table 1—Summary of rock types and induced fracture orientation.

Sample ID	Rock type	Orientation
IND1	Carbonate rocks, which are rich in calcite minerals, exhibit a single porosity model according to NMR measurements.	Horizontal fracture orientation, where the sample was subjected to hydraulic fracturing at one-third of its inlet surface.
IND2		Vertical orientation, where the sample was subjected to hydraulic fracturing at the midpoint.
BND1	Sandstone formations, composed of various minerals such as quartz and clay minerals, exhibit dual porosity models and are more tightly packed compared to carbonate rocks, as shown by NMR measurements.	Horizontal fracture orientation, where the sample was subjected to hydraulic fracturing at one-third of its inlet surface.
BND2		The sample was subjected to hydraulic fracturing in a vertical orientation, with the fracture induced at the midpoint.

Experimental Procedure

Two experimental setups were employed to measure dielectric properties (permittivity and conductivity) and nuclear magnetic resonance response. Figure 1A illustrates the open-ended coaxial probe system used to acquire the dielectric permittivity and conductivity profiles. Prior to hydraulic fracturing, measurements were taken at 40 distinct frequencies for both dry and saturated samples. After fracturing, the number of data points was increased to 80 to improve spatial resolution and accuracy; however, the full frequency range (1 MHz to 3 GHz) was covered in all measurements. For each core sample, measurements were collected at two locations, one near the inlet and another near the outlet. A total of 16 measurement sets across all samples have been obtained that cover two rock types (carbonate and sandstone), two fracture orientations (horizontal and vertical perpendicular), two distances (one-third and two-thirds), and two sample conditions (dry and brine saturated). More data points were obtained after applying the fracturing treatment in order to better capture the heterogeneity introduced by the induced fractures. Figure 1B shows the GeoSpec NMR instrument, which was utilized to measure both incremental and cumulative porosity of the saturated samples. T_2 relaxation profiles were obtained to provide complementary insights into pore structure and fluid distribution.



Figure 1—Experimental setups used for dielectric and NMR measurements. (A) Open-ended coaxial reflection probe calibrated with standard materials for accurate dielectric permittivity and conductivity measurements. (B) Oxford Instruments GeoSpec2-75 NMR system used to evaluate incremental and cumulative porosity in saturated rock samples.

Results and Discussion

This section discusses the experimental findings obtained from dielectric and NMR measurements conducted on Indiana limestone and Bandera sandstone samples with different fracture orientations (as outlined in **Table 1**). The data reflect the distinct electromagnetic and pore-scale responses of the rock samples under dry and saturated conditions. For both rock types, measurements were carried out before and after fluid saturation, allowing for a direct comparison of how hydraulic fracturing and fluid presence influence dielectric permittivity, conductivity, and NMR profiles. The saturated condition, achieved by injecting 3 wt.% KCl brine into the core samples under a pressure of 1900 psi, revealed notable contrasts in dielectric response compared to the dry state. These differences highlight the combined influence of fracture orientation, lithology, and measurement location (inlet vs. outlet) on the observed dielectric. The following subsections discuss the trends observed across different configurations, focusing on the impact of fluid presence, fracture geometry, and lithological characteristics on the electrical and NMR behaviors.

Nuclear Magnetic Resonance Analysis

The primary objective of using NMR measurement is to gain more insight into the pore properties of the samples before and after inducing the fractures. NMR results represent the change of incremental and cumulative porosity versus relaxation time T_2 , which measures the decay of the magnetization. T_2 profiles are significantly influenced by the interactions between water molecules and the surrounding pore walls. Smaller pores lead to faster T_2 relaxation due to increased surface interactions, while larger pores exhibit slower relaxation. This relationship allows us to assess the pore size distribution from T_2 relaxation measurements. Both rock types (carbonate and sandstone) were measured under the condition of saturation. In total, there are four separate measurements were taken, each representing an orientation and type of rock. In those four measurements, incremental and cumulative porosity (ϕ) were calculated versus relaxation time (T_2).

The incremental porosity curve, which represents the distribution of pore sizes, shifted downwards after inducing the fracture. This shift indicates less water volume in the large pores inside the sample, which mainly forms the fracture body. Although the water loss was minimized during the NMR measurement by covering the samples with Teflon/nylon, but still some reduction in the total porosity was still observed. Such conditions could be resolved in the future by running the measurements in a confined and high-pressure system. This reduction is likely due to the smaller sample size. Furthermore, the incremental porosity curve is centered around 200 ms with one noticeable peak, suggesting that IND1 has only a single porosity. To summarize, the analysis of T_2 relaxation times via T_2 NMR provided valuable insight into the pore size distribution within rock samples and how this distribution is affected by hydraulic fracturing. The decrease

in cumulative porosity highlights a reduction in total pore volume, likely due to the reduced water volume. Furthermore, the single peak indicates that IND1 exhibits a single porosity model, which is indicative of its uniformity.

Figure 2 illustrates the effect of hydraulic fracturing on the NMR response of the carbonate sample IND1. The figure presents both incremental and cumulative porosity as functions of T_2 relaxation time, enabling a comparison between pre- and post-fracture conditions. For the horizontally fractured IND1 sample, the cumulative porosity decreased from 18 to 16.6 p.u., indicating a reduction in the water-filled pore volume. However, the incremental porosity curve showed a downward shift after fracturing, which reflects the changes in the pore size distribution. This suggests a decrease in water volume, particularly in the larger pores that constitute the fracture body. Although the samples were wrapped in nylon to minimize fluid loss during NMR measurements, some reduction in total porosity was still observed, likely due to exposure to ambient conditions and the smaller post-fracture sample volume. To mitigate such effects, future measurements could be performed under confined and high-pressure conditions to better preserve fluid saturation. Additionally, the incremental porosity curve remained centered around 200 ms, exhibiting a single dominant peak, which indicates that IND1 has a single-porosity system with a relatively homogeneous pore structure. In summary, the T_2 NMR analysis revealed that hydraulic fracturing led to a measurable reduction in cumulative porosity, primarily due to decreased water content in large pore spaces. The presence of a single, well-defined peak in the T_2 distribution supports the interpretation that IND1 has a uniform pore structure, consistent with a single-porosity model.

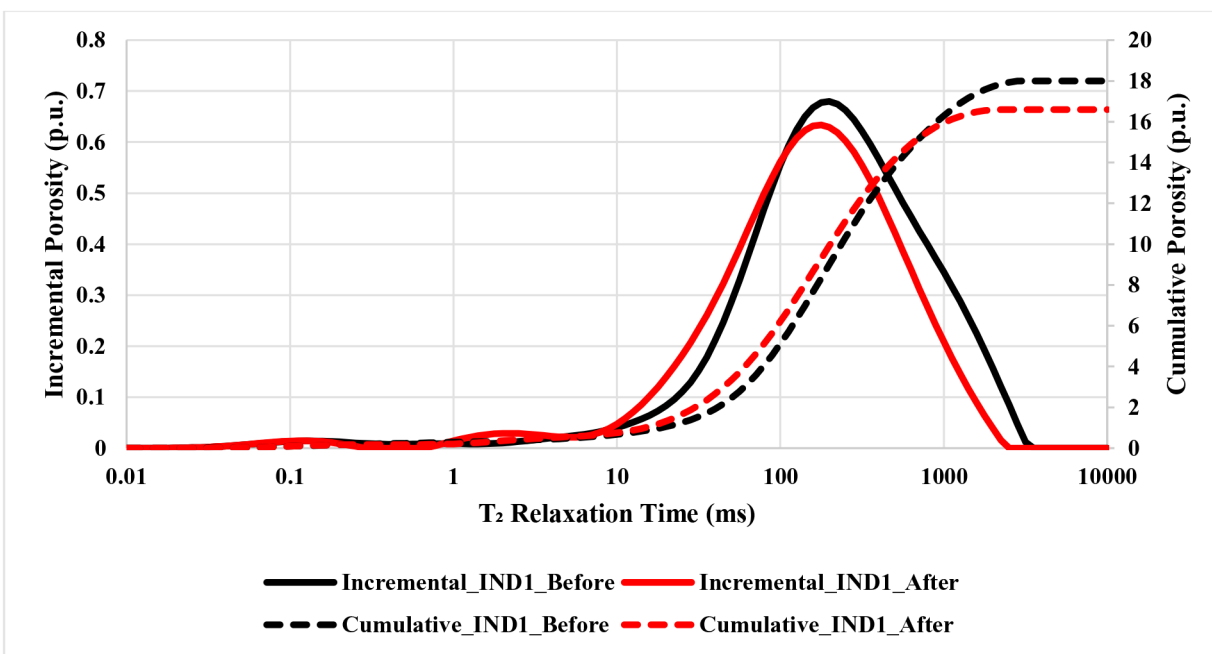


Figure 2— T_2 NMR response of the brine-saturated carbonate sample (IND1) before and after hydraulic fracturing. The red solid curve shows the incremental porosity, representing pore size distribution. The red dotted curve indicates cumulative porosity versus T_2 , illustrating the total fluid-filled pore volume.

Figure 3 illustrates the effect of hydraulic fracturing on the T_2 NMR response of the carbonate sample IND2. Both incremental and cumulative porosity curves are plotted as functions of T_2 relaxation time, providing insight into the pore size distribution and total fluid-filled pore volume before and after inducing the fracture. Sample IND2 was subjected to a vertical fracture, unlike IND1, which was fractured horizontally. This change in orientation not only altered the fracture geometry but also influenced fluid distribution and surface exposure. While the vertical cut reduced the overall volume of the sample, it significantly increased the exposed internal surface area, which may have promoted more brine retention in

finer pore structures and along fracture surfaces. After HF, the incremental porosity curve shifted downward, indicating a reduction in overall water volume, particularly within larger pores. This suggests that some of the larger connected pore pathways may have been compromised or partially drained, while smaller pore regions retained the brine. Overall, the incremental porosity curve exhibits no noticeable change in shape before and after fracturing, indicating that the internal pore architecture remained largely unaffected by the HF process. The cumulative porosity was decreased from 18 to 16.7 p.u., consistent with the trend observed in IND1. This reduction is largely attributed to water loss during NMR experiments, despite efforts to minimize evaporation by wrapping the samples in nylon. The similarity in porosity reduction carbonates reinforces that handling-related water loss is a dominant factor affecting the measurements, rather than the fracture orientation alone. In summary, the T_2 response of IND2 shows that vertical hydraulic fracturing alters pore structure and fluid distribution in a way that slightly differs from horizontal fracturing. Although total porosity decreased post-fracture, the subtle differences in the incremental profile suggest that fracture orientation plays a role in modifying the internal connectivity and brine retention behavior of carbonate rocks.

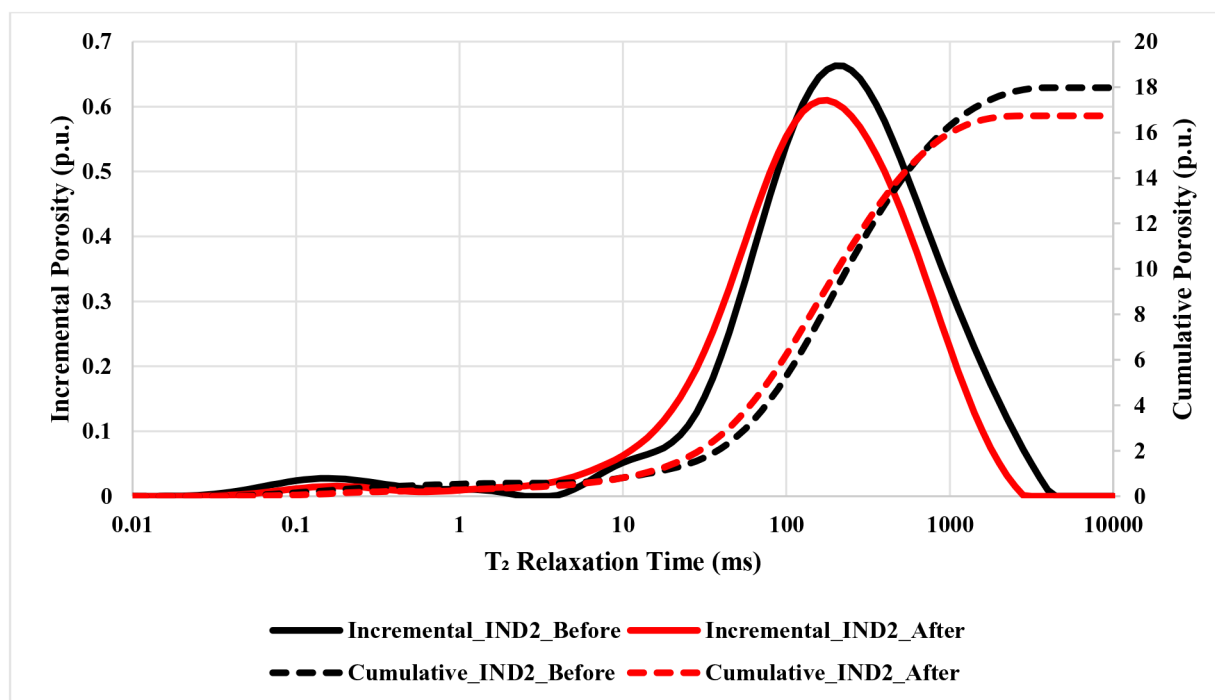


Figure 3— T_2 NMR response of the second brine-saturated carbonate sample (IND2) before and after hydraulic fracturing. The solid curves represent incremental porosity as a function of T_2 relaxation time, indicating pore size distribution. The dotted curves show cumulative porosity versus T_2 , reflecting the total fluid-filled pore volume.

NMR profiles were obtained for the sandstone samples both before and after hydraulic fracturing. Figure 4 presents the results for sample BND1, which was horizontally fractured, while Figure 5 shows data for the sample fractured vertically. Initially, both samples exhibit a dominant T_2 peak around 70 ms, accompanied by two smaller secondary peaks near 2 ms and 0.2 ms, indicating the presence of multiple pore systems. The cumulative porosities for the two samples were 19.5 and 20 p.u., respectively. Following hydraulic fracturing, only minimal changes were observed in the NMR profiles. For the horizontally fractured sample (BND1), total porosity decreased from 19.5 to 17.7 p.u., whereas the vertically fractured sample showed almost no change in cumulative porosity. When comparing NMR results across both carbonate and sandstone samples, it appears that fracture orientation cannot be distinctly resolved by the T_2 NMR technique. The observed reductions in porosity primarily reflect water loss from the fracture zone rather than

structural changes in the pore network. Since T_2 NMR provides a bulk evaluation of the core, more advanced NMR methods may be necessary to detect subtle variations caused by different fracture orientations. The next phase of this study involves performing multifrequency dielectric measurements on the same samples to further investigate how fracture presence and orientation affect dielectric properties.

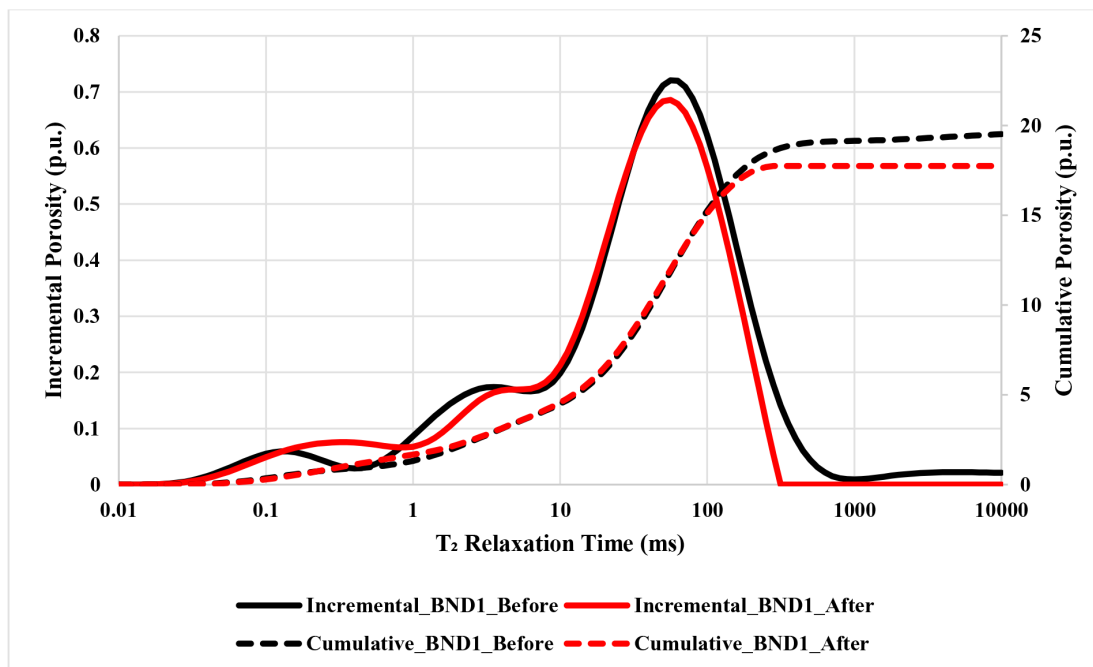


Figure 4— T_2 NMR profile of the brine-saturated sandstone sample (BND1) before and after hydraulic fracturing. The solid curves depict incremental porosity, highlighting the pore size distribution. The dotted curves represent cumulative porosity versus T_2 , indicating the total fluid-filled pore volume within the sample pre- and post-fracture.

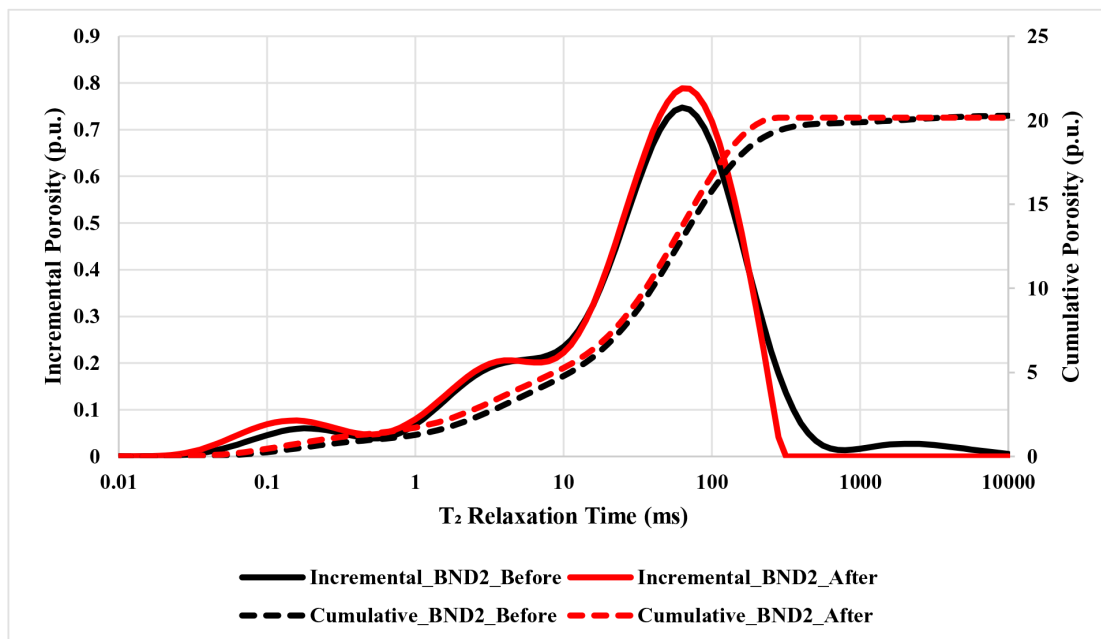


Figure 5— T_2 NMR profile of the second brine-saturated sandstone sample (BND2) before and after hydraulic fracturing. The incremental porosity is represented by the solid curves, while the cumulative porosity is depicted by the dotted curves.

Analysis of Dielectric Behavior for Rock Matrix

The primary objective of measuring the dielectric properties of the samples is to evaluate the influence of fracture presence and orientation on the dielectric behavior of carbonate and sandstone formations. Both relative permittivity and electrical conductivity profiles were determined under various conditions. Initial measurements were conducted on dry samples to characterize the dielectric response of the rock matrix, independent of fluid interactions. [Figure 6](#) presents the relative permittivity profiles for the carbonate samples before and after hydraulic fracturing, including both horizontal and vertical fracture orientations. The measured permittivity values ranged between 5.3 and 7 (dimensionless), aligning well with values reported in the literature for dry carbonate rocks. A slight decrease in permittivity with increasing frequency was observed, likely due to residual moisture. However, no significant difference was found between the pre- and post-fracture profiles, suggesting that the fractures do not significantly alter the bulk dielectric response in the absence of brine. This is attributed to the lack of interactive fluids within the porous network, meaning no interfacial polarization mechanisms were active. Moreover, the corresponding conductivity profiles for the carbonate samples are shown in [Figure 7](#). For all samples, conductivity remained below 0.02 S/m, confirming the effective dryness of the samples. The conductivity trends were nearly identical before and after fracture induction, further indicating that fracture presence and orientation cannot be assessed through dielectric measurements in dry samples. These results also confirm that the rock matrix remains structurally intact post-fracturing, at least from an electrical perspective. Similar trends were observed for the sandstone samples, as illustrated in [Figures 8 and 9](#). Both permittivity and conductivity profiles showed negligible variation after hydraulic fracturing, regardless of fracture orientation. These findings collectively demonstrate that, under dry conditions, the dielectric properties of both carbonate and sandstone samples are primarily governed by the matrix composition, with minimal sensitivity to induced fractures due to the absence of interfacial polarization.

In conclusion, the dry-state dielectric measurements indicate no significant alteration in matrix permittivity or conductivity due to fracturing. The absence of brine results in a lack of polarization mechanisms, which are essential for detecting interfacial effects and distinguishing the presence and orientation of fractures. Therefore, to better assess fracture impacts and enable interfacial polarization phenomena, dielectric measurements were also performed on brine-saturated samples, which will be discussed in the following sections.

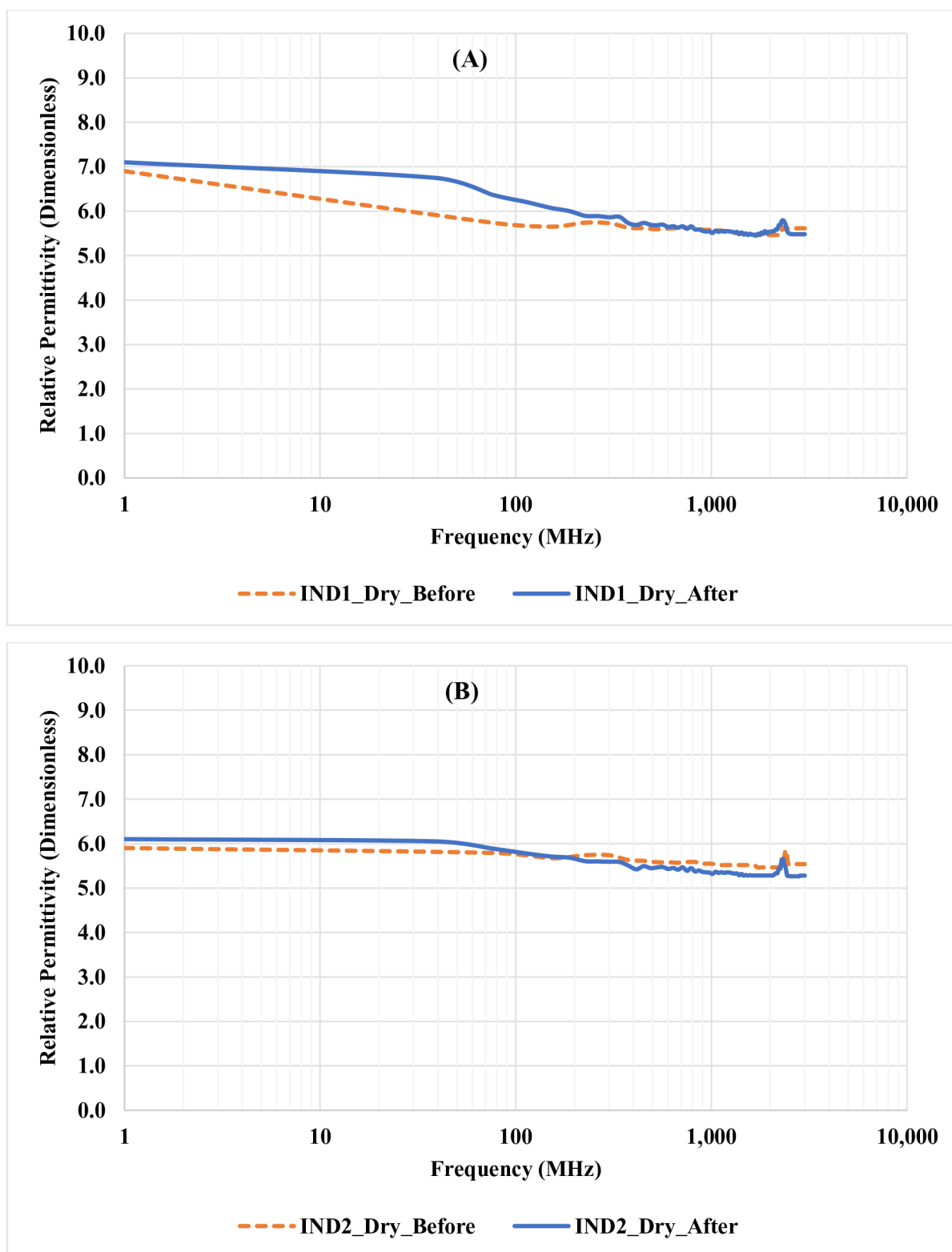


Figure 6—Relative permittivity profiles of dry carbonate samples before and after hydraulic fracturing. (A) sample IND1 with a horizontal fracture, and (B) Sample IND2 with a vertical fracture. The plots compare the dielectric response of the carbonate samples under dry conditions, highlighting the effect of fracture orientation on relative permittivity.

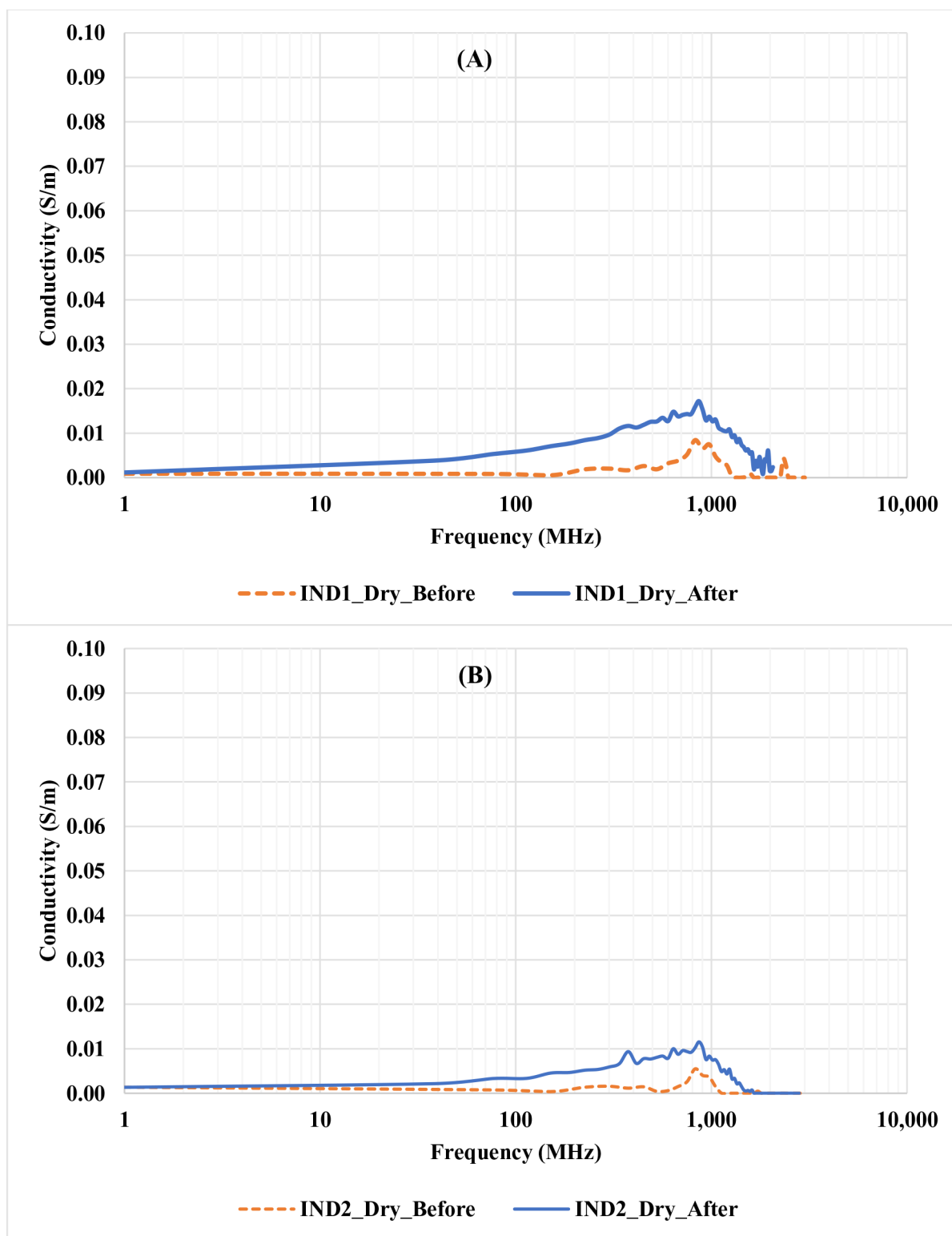


Figure 7—Electrical conductivity profiles of dry carbonate samples before and after hydraulic fracturing. (A) Sample IND1 with a horizontal fracture, and (B) Sample IND2 with a vertical fracture. The plots show the conductivity response under dry conditions, indicating minimal impacts for fracture presence or orientation.

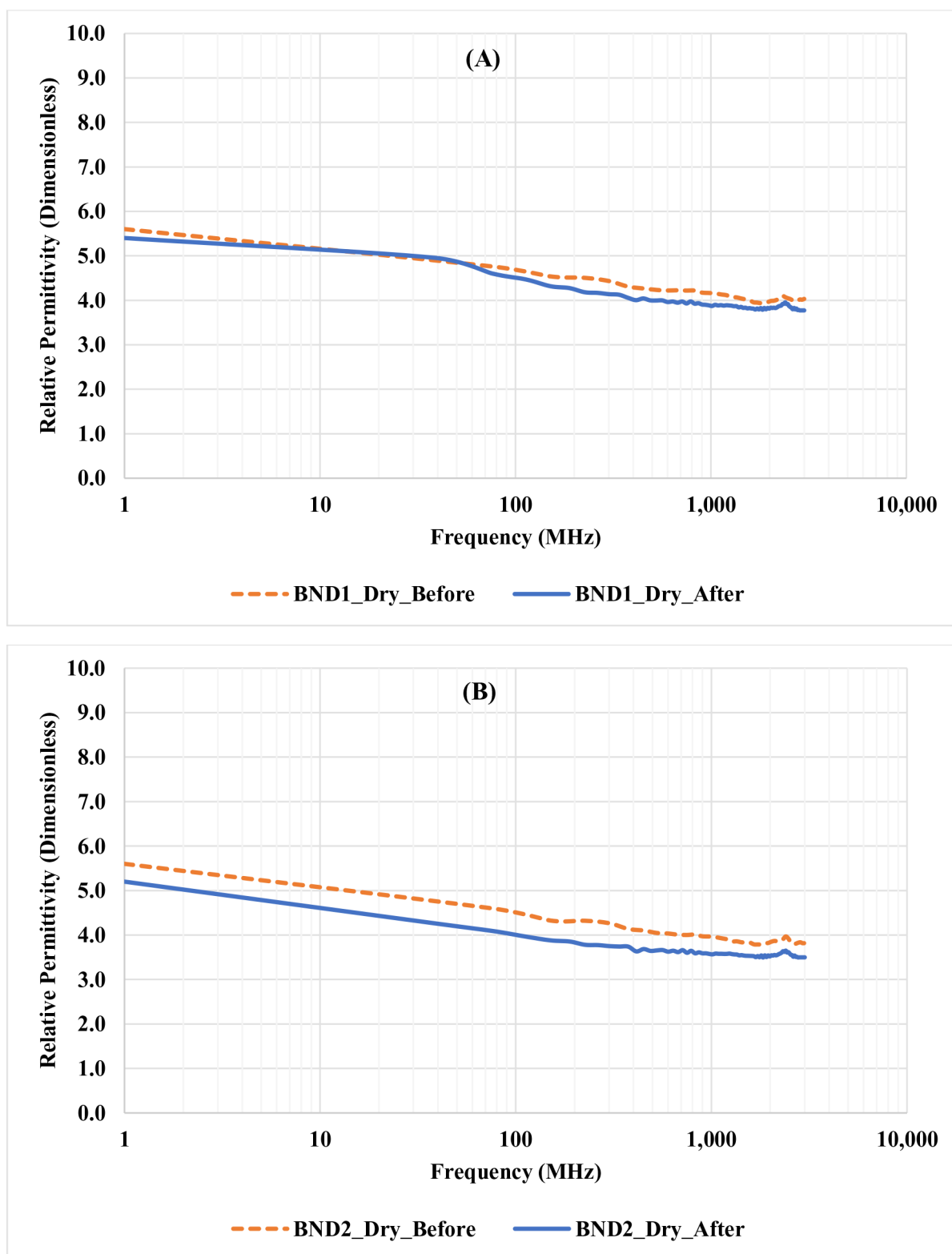


Figure 8—Relative permittivity profiles of dry sandstone samples before and after hydraulic fracturing. (A) Sample BND1 with a horizontal fracture, and (B) Sample BND2 with a vertical fracture. The plots compare the dielectric response of the sandstone samples under dry conditions, highlighting the effect of fracture orientation on relative permittivity.

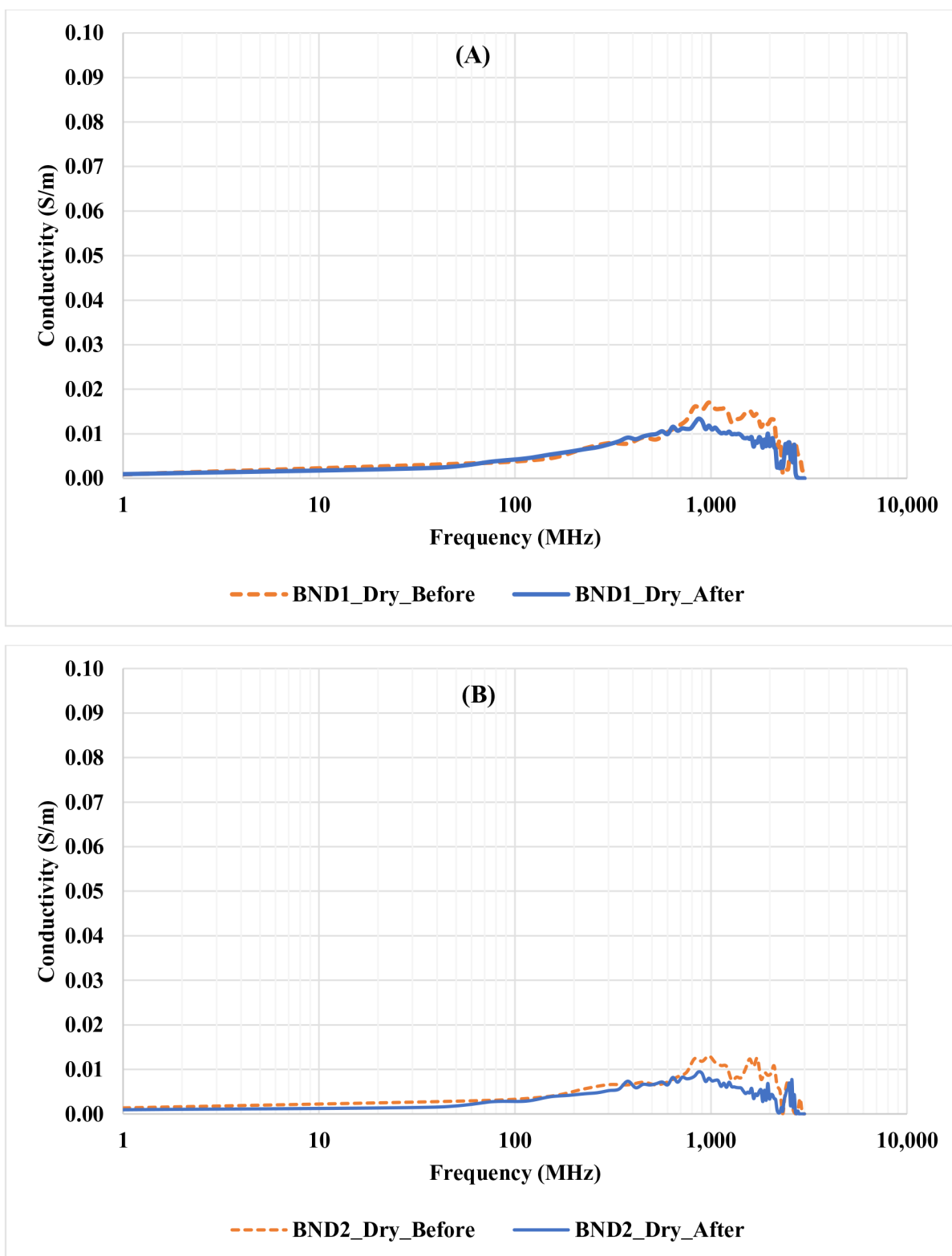


Figure 9—Electrical conductivity profiles of dry sandstone samples before and after hydraulic fracturing. (A) Sample BND1 with a horizontal fracture, and (B) Sample BND2 with a vertical fracture. The plots show the conductivity response under dry conditions, indicating minimal impacts for fracture presence or orientation.

Dielectric Dispersion Analysis of Brine-Saturated Rocks

Similar to the dry-state measurements, the dielectric properties of all four carbonate and sandstone samples were evaluated using the open-ended coaxial probe technique under saturated conditions. [Figure 10](#) presents the relative permittivity profiles of the brine-saturated carbonate samples before and after hydraulic fracturing, with fracture orientations being horizontal for sample IND1 and vertical for sample IND2. These profiles capture the frequency-dependent dielectric response, highlighting the effect of fracture presence and geometry on permittivity behavior. At low frequencies, the relative permittivity increased significantly after fracturing, attributed to enhanced interfacial polarization caused by the increased surface area along the fracture faces. Specifically, the permittivity was increased from 45 to 70 for IND1 and from 49 to 68 for IND2 following fracture induction. In contrast, at higher frequencies, the permittivity values before and after fracturing remained nearly identical, indicating that total porosity was not substantially changed and the dielectric response at these frequencies is dominated by the matrix properties and brine volume. The difference between the pre- and post-fracture permittivity curves was more pronounced in IND2, which experienced vertical fracturing. This is consistent with the larger exposed fracture surface area in the vertical orientation of around 9 in², compared to approximately 3.54 in² for the horizontally fractured IND1. These estimates are based on sample dimensions of 1.5 inches in diameter and 3 inches in length, where the vertical fracture was induced along the length of the core and the horizontal fracture was induced across the diameter. These observations confirm that under saturated conditions, fracture orientation and surface area can strongly influence the dielectric response at lower frequencies due to interfacial effects, while having minimal impact at higher frequencies where matrix properties dominate.

The electrical conductivity profiles of brine-saturated carbonate samples before and after hydraulic fracturing with different fracture orientations are presented in [Figure 11](#). [Figure 11A](#) corresponds to sample IND1 with a horizontal fracture, while [Figure 11B](#) shows data for sample IND2 with a vertical fracture. The plots depict the frequency-dependent conductivity response under saturated conditions, emphasizing the impact of fracture geometry on the effective electrical conductivity of carbonate rocks. Following the hydraulic fracturing, an increase in conductivity was observed in both samples. This increase is more pronounced in IND2, the vertically fractured sample, due to its longer connected fluid pathway and larger exposed surface area, which can facilitate enhanced ionic transport through the brine-filled fracture. The horizontal fracture in IND1 resulted in a smaller increase in conductivity, which is consistent with its shorter conductive pathway. An additional contributing factor may be the probe placement relative to the fracture geometry. In IND2, the vertical fracture extends from the core inlet to the outlet, meaning the fracture likely remains in direct contact with the coaxial probe during measurement. This direct connection provides a continuous brine-filled path for current flow, enhancing the measured conductivity. In contrast, the horizontal fracture in IND1 does not intersect the probe location, and therefore, the probe does not directly contact the conductive fluid path. As a result, the effect of the fracture on conductivity is less pronounced in IND1, further emphasizing the role of fracture-probe alignment and fluid connectivity in determining the effective electrical response.

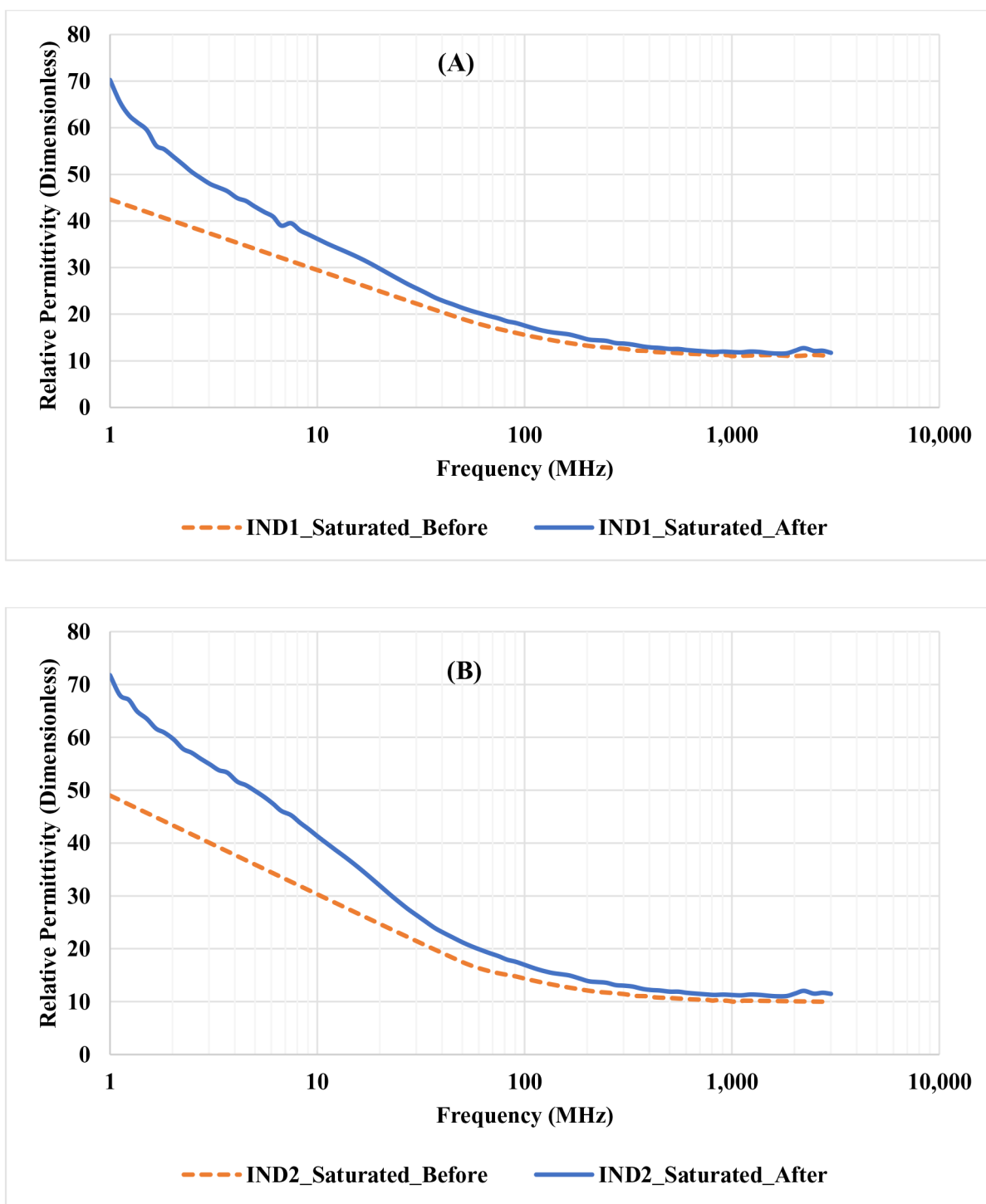


Figure 10—Relative permittivity profiles of brine-saturated carbonate samples before and after hydraulic fracturing with different fracture orientations. (A) Sample IND1 with a horizontal fracture, and (B) Sample IND2 with a vertical fracture. The profiles illustrate the frequency-dependent dielectric response under saturated conditions, highlighting the influence of fracture geometry on permittivity behavior.

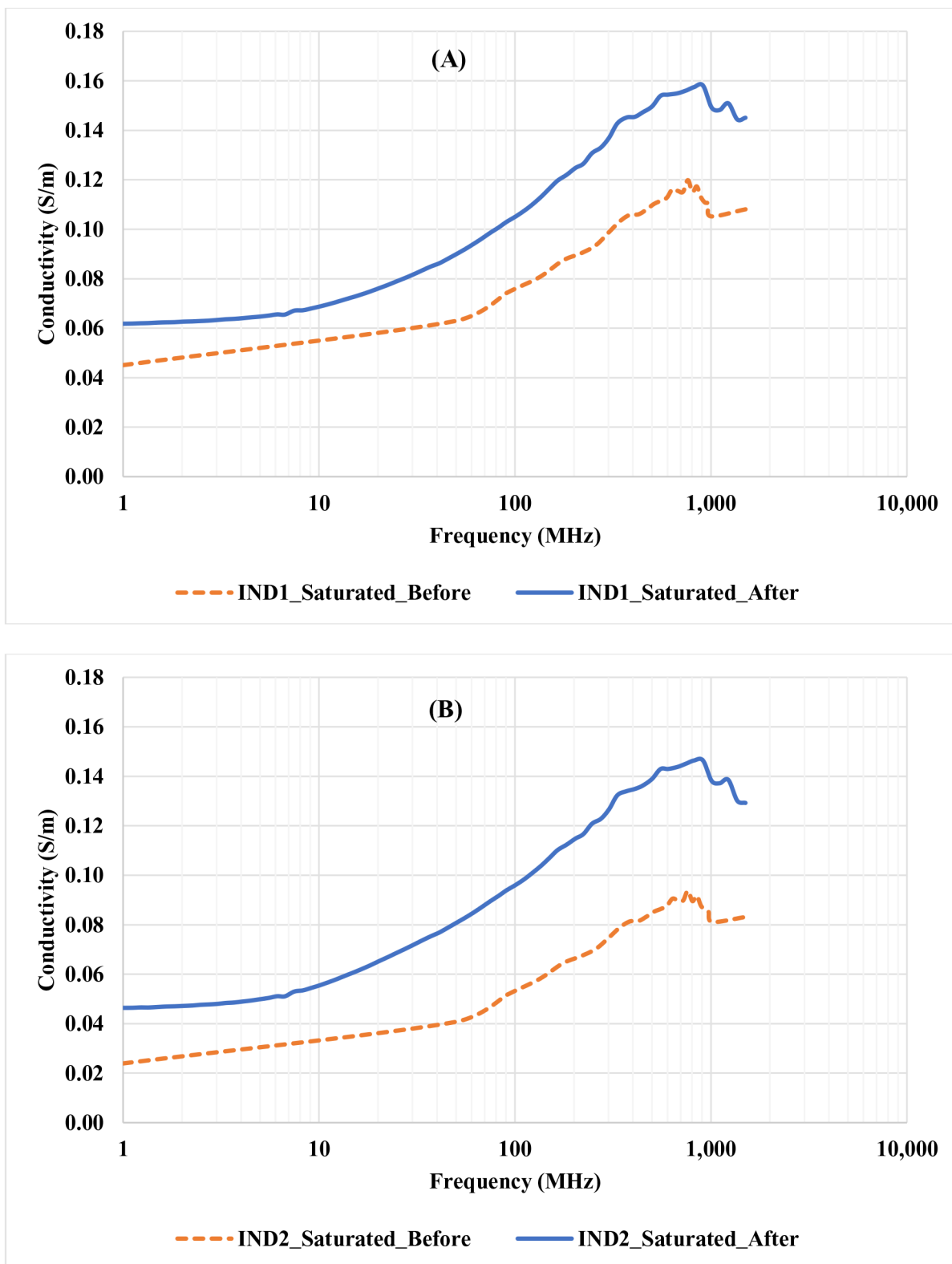


Figure 11—Electrical conductivity profiles of brine-saturated carbonate samples before and after hydraulic fracturing with different fracture orientations. (A) Sample IND1 with a horizontal fracture, and (B) Sample IND2 with a vertical fracture. The plots illustrate the frequency-dependent conductivity response under saturated conditions, highlighting the influence of fracture geometry on the effective electrical conductivity for carbonate rocks.

In addition, the relatively smooth trends in the conductivity profiles suggest minimal fracture roughness, which aligns with physical observations of the tested samples. However, it is important to note that in real reservoir conditions, fracture surfaces are often rougher and more irregular, potentially impeding fluid flow and reducing conductivity. Nevertheless, the conductivity in such cases would still be expected to remain higher than or equal to the baseline (pre-fracture) conductivity due to the presence of brine and increased surface area. The presence of different fluids (oil and brine) in the matrix and fracture body would induce more complexity, which needs further studies. In summary, the conductivity results confirm that under saturated conditions, both fracture orientation and geometry play a significant role in enhancing electrical conductivity, with vertical fractures producing a greater effect due to extended conductive pathways.

Moreover, the relative permittivity profiles of brine-saturated sandstone samples before and after hydraulic fracturing with different fracture orientations are presented in [Figure 12](#), while the corresponding electrical conductivity profiles are shown in [Figure 13](#). These plots illustrate the frequency-dependent dielectric response under saturated conditions, highlighting the influence of fracture geometry on both permittivity and conductivity in sandstone formations. Overall, the dielectric behavior of the sandstone samples followed a trend similar to that observed in the carbonate rocks; the presence of hydraulic fracturing resulted in an increase in both relative permittivity and electrical conductivity. This enhancement is primarily attributed to the larger surface area introduced by the fractures, which promotes interfacial polarization and improves ionic conduction through the brine-saturated pathways. However, unlike the carbonate samples, the difference between vertical and horizontal fracture orientations in sandstone is less conclusive. This ambiguity may be related to the complex mineralogy of sandstone, particularly the presence and spatial distribution of conductive minerals and clay, which can mask the direct effects of fracture geometry on the measured dielectric response. To better interpret these results, additional analyses are recommended, specifically by characterizing the location and concentration of conductive minerals near the fractures and at the measurement points. Such investigations could help isolate the contribution of mineralogical variability from the effects of fracture orientation. In summary, the presence of hydraulic fractures in sandstone generally enhances both permittivity and conductivity due to increased surface area, but mineralogical heterogeneity may limit the ability to clearly distinguish the impact of fracture orientation alone.

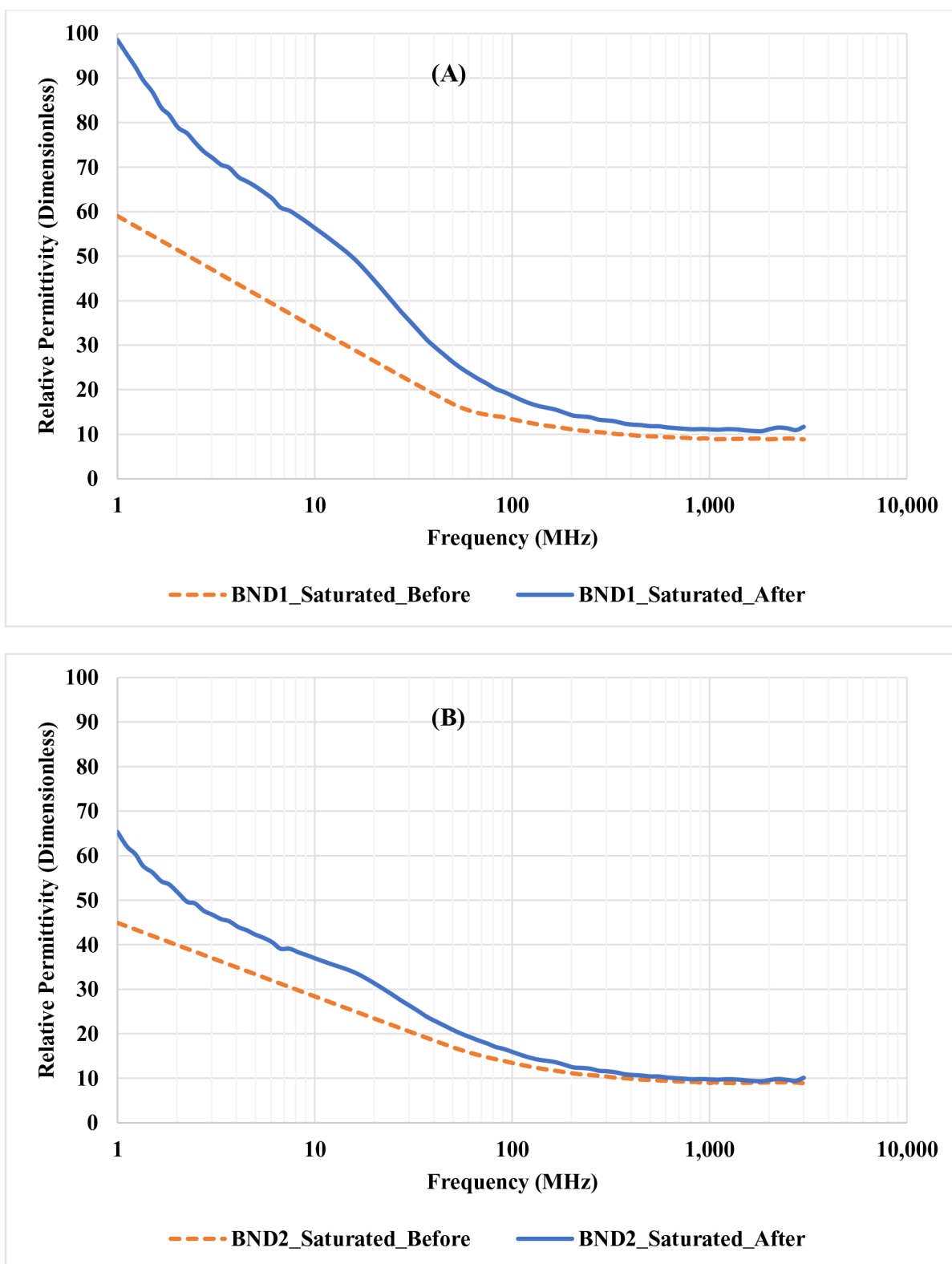


Figure 12—Relative permittivity profiles of brine-saturated sandstone samples before and after hydraulic fracturing with different fracture orientations. (A) Sample BND1 with a horizontal fracture, and (B) Sample BND2 with a vertical fracture. The profiles illustrate the frequency-dependent dielectric response under saturated conditions, highlighting the influence of fracture geometry on permittivity behavior.

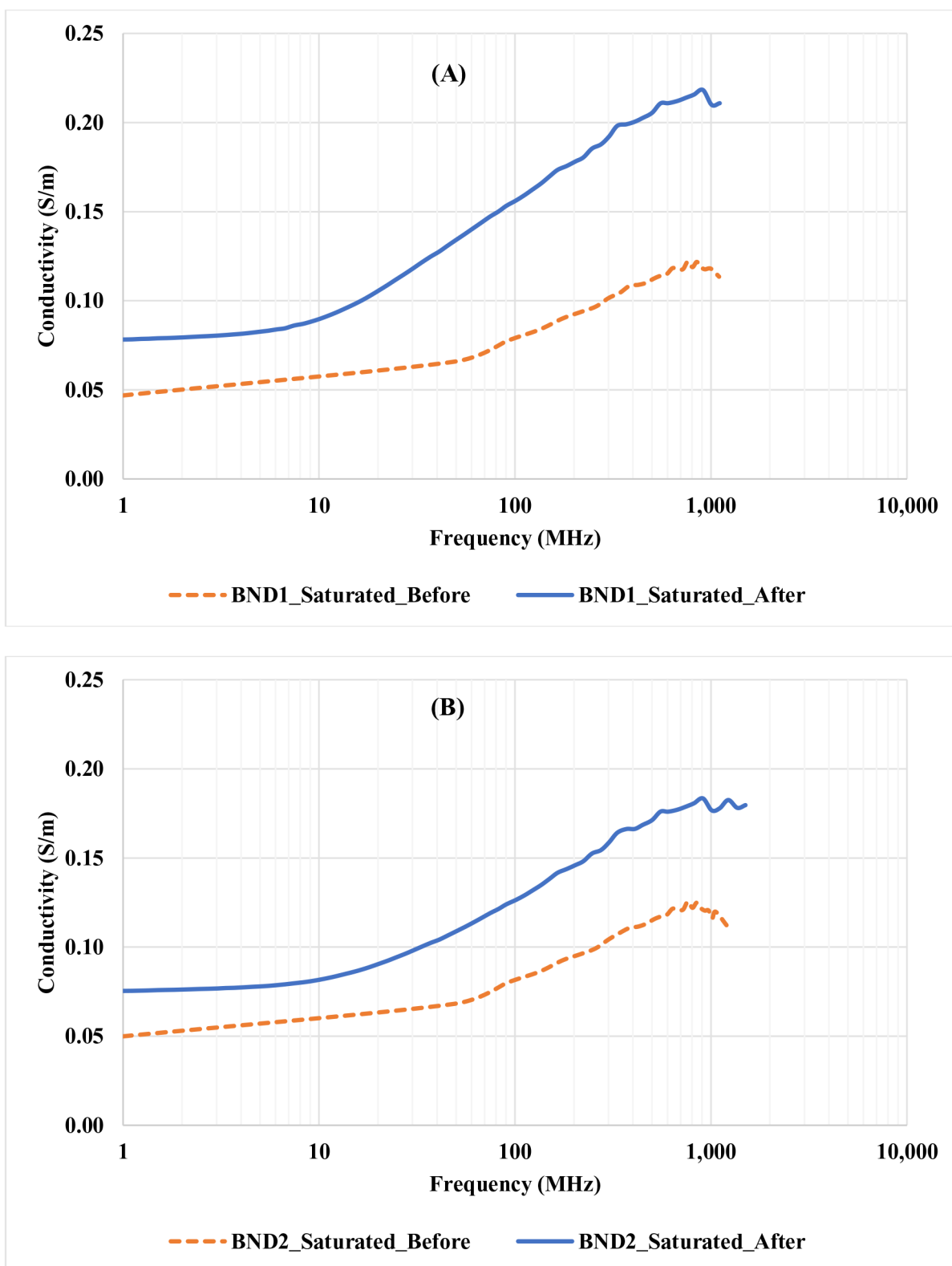


Figure 13—Electrical conductivity profiles of brine-saturated sandstone samples before and after hydraulic fracturing with different fracture orientations. (A) Sample BND1 with a horizontal fracture, and (B) Sample BND2 with a vertical fracture. The plots illustrate the frequency-dependent conductivity response under saturated conditions, highlighting the influence of fracture geometry on the effective electrical conductivity for sandstone rocks.

Ultimately, the presence of hydraulic fractures can increase both relative permittivity and conductivity in brine-saturated carbonate and sandstone samples due to the larger internal surface area and better fluid connectivity. In carbonates, fracture orientation had a clear impact, where the vertical fractures produced a greater dielectric response, likely due to longer conductive paths and direct probe contact. In contrast, sandstone samples showed less sensitivity to fracture orientation, likely due to their complex mineralogy and the presence of conductive clays. Overall, the multifrequency dielectric technique can effectively capture the presence and orientation of fractures, and its ability to distinguish fracture orientation is more reliable in carbonates than in sandstones.

Impact of Fracture Distance on Dielectric Dispersions

This section examined how varying the distance between the dielectric probe and the fracture body influences permittivity and conductivity. [Figure 14](#) presents the permittivity and conductivity profiles of a carbonate sample measured at two positions relative to the induced fracture. The "near" measurement corresponds to a probe location approximately 1 inch from the fracture, while the "far" measurement is taken at a distance of 2 inches, within a total core length of 3 inches. The relative permittivity measured near the fracture was slightly higher than that measured farther away. This behavior is attributed to the dielectric penetration depth of approximately 2–4 inches, which allows the probe to sense both the fracture and the surrounding matrix. However, in general, proximity to the fracture increases the relative permittivity, particularly at low frequencies, due to the high permittivity of brine filling the fracture. In contrast, the electrical conductivity exhibited a more pronounced difference between near and far measurements. At 1 GHz, the near-fracture location showed an approximate 34% increase in conductivity, indicating improved ionic connectivity and confirming the localized effect of the brine-filled fracture on the dielectric response. On the other hand, the dielectric response of the sandstone sample was evaluated at varying distances from the induced fracture, as shown in [Figure 15](#). The relative permittivity exhibited a noticeable increase at low frequencies (below 10 MHz) when measured near the fracture, indicating enhanced interfacial polarization due to the brine-filled fracture zone. Additionally, the electrical conductivity increased by approximately 12% at the near-fracture location compared to the far position, highlighting improved ionic conduction along the fluid-filled fracture pathway.

Overall, the multifrequency dielectric technique demonstrated a significant sensitivity to fracture distance. Both relative permittivity and electrical conductivity profiles were notably higher near the fracture compared to measurements taken farther away in carbonate and sandstone samples. These results underscore the strong local influence of fracture proximity on the dielectric response, suggesting its potential usefulness for estimating fracture distance. Furthermore, a correlation could be developed to predict fracture location based on formation matrix properties, formation fluid characteristics, and dielectric responses.

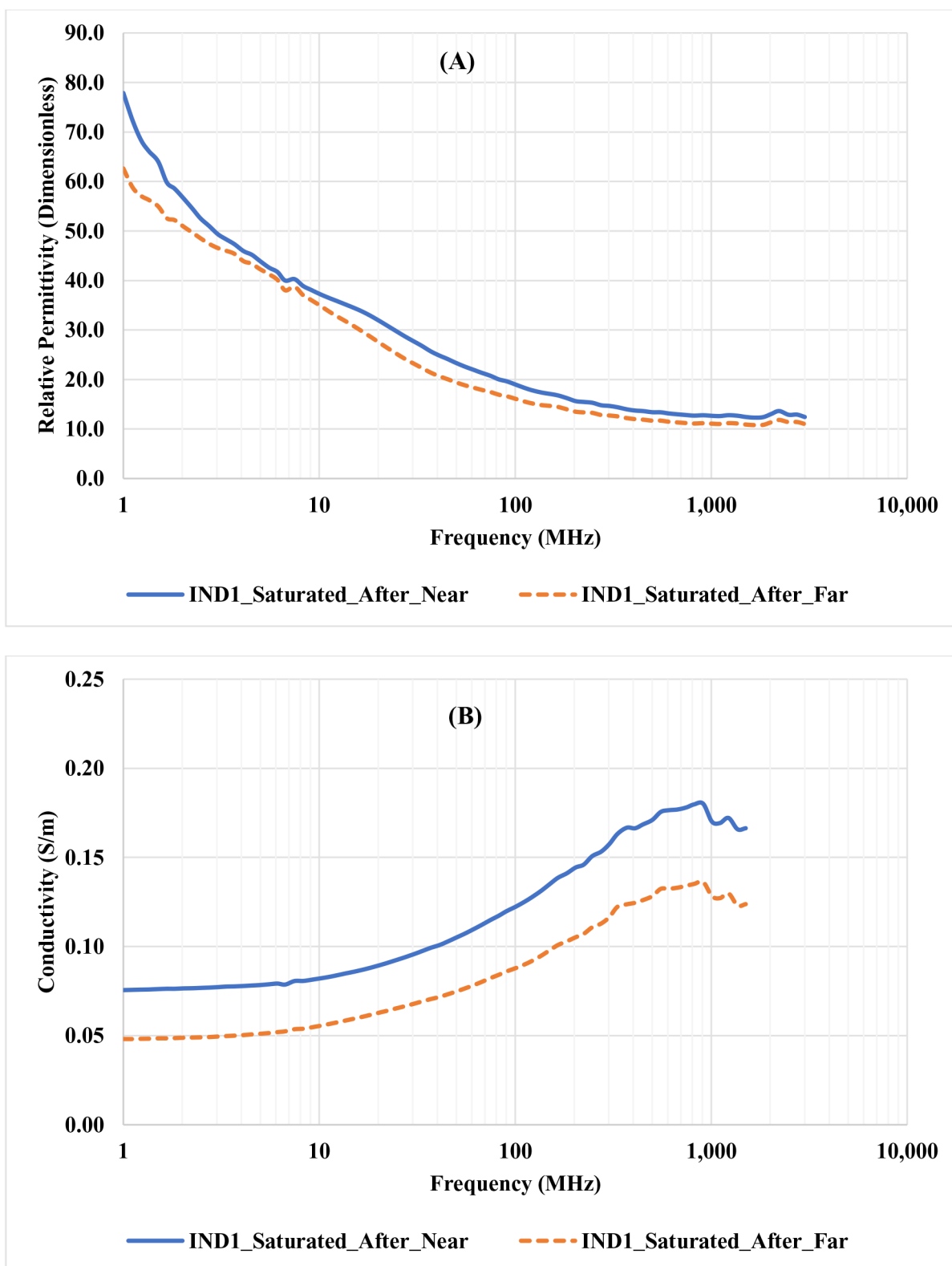


Figure 14—Relative permittivity (A) and electrical conductivity (B) profiles of a brine-saturated carbonate sample measured at two distances from the induced fracture. The plots highlight the influence of fracture proximity on the dielectric response in carbonate formations.

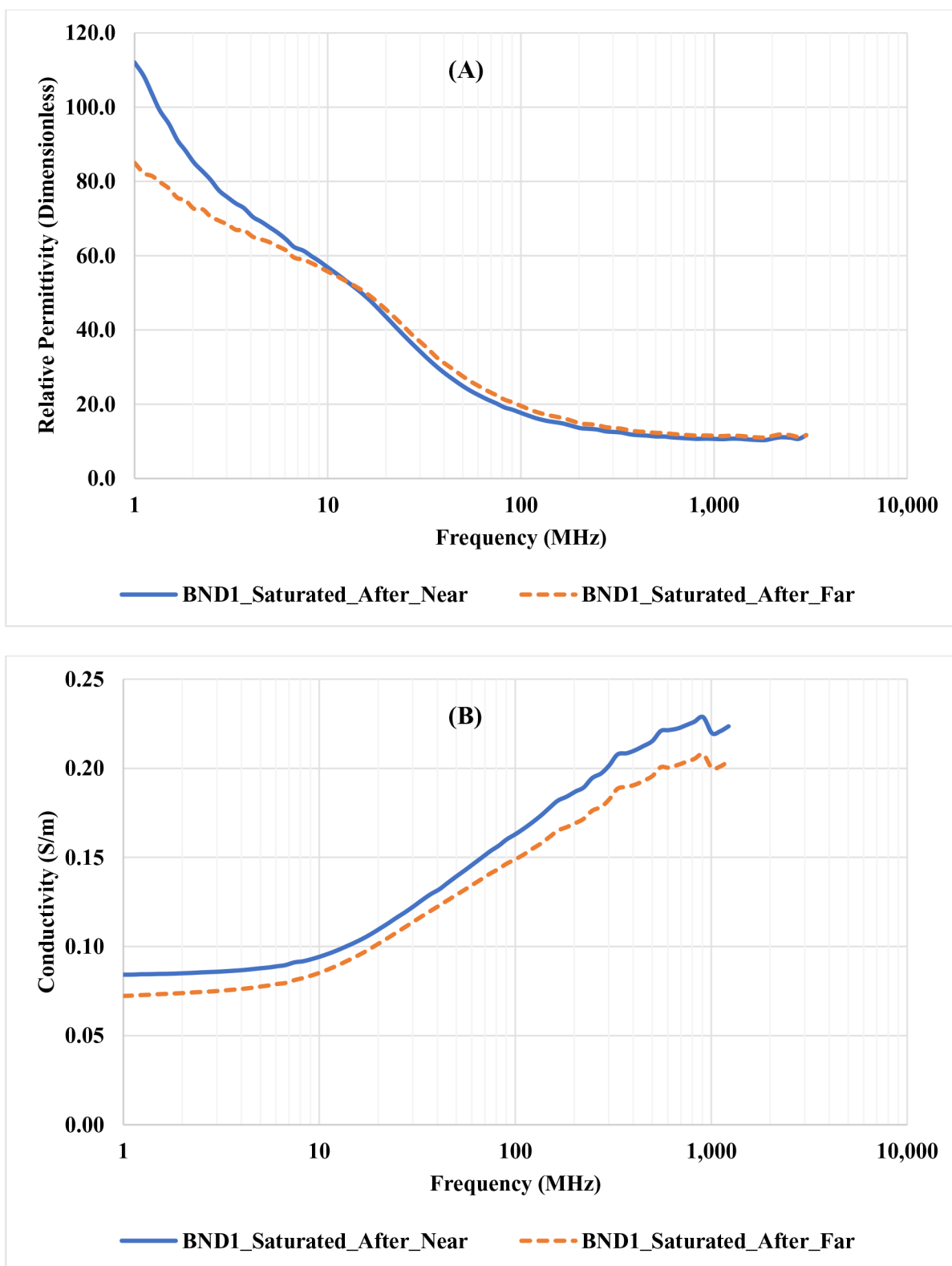


Figure 15—Relative permittivity (A) and electrical conductivity (B) profiles of a brine-saturated sandstone sample measured at two distances from the induced fracture. The plots highlight the influence of fracture proximity on the dielectric response in sandstone formation.

Conclusions

This study investigates the impact of fracture presence, orientation, and proximity on the dielectric response of rock samples under dry and brine-saturated conditions. Multifrequency dielectric and T_2 NMR measurements were conducted. The experiments were performed on carbonate and sandstone samples, with fractures induced in two orientations, vertical and horizontal. Based on the results obtained, the following conclusions can be drawn:

- T_2 NMR profiles indicated the presence of fractures but did not clearly differentiate between fracture orientations; this suggests that more advanced NMR techniques are required to detect subtle variations associated with fracture geometry.
- Dry-state dielectric measurements showed no significant changes in relative permittivity or conductivity after fracturing, due to the absence of brine and resulting lack of polarization mechanisms necessary for detecting interfacial effects or fracture orientation.
- In brine-saturated conditions, hydraulic fractures led to increased relative permittivity and electrical conductivity in both carbonate and sandstone samples, driven by enhanced fluid connectivity and increased internal surface area along the fracture paths.
- Fracture orientation and proximity significantly influenced the dielectric response in carbonates, with vertical fractures showing a stronger effect due to longer conductive paths and direct probe interaction. Sandstone samples exhibited less pronounced orientation effects, likely due to their complex mineralogy and clay content.
- Overall, the multifrequency dielectric technique proved effective in detecting the presence, proximity, and orientation of fractures with more reliability in carbonates. The obtained results can support the development of predictive correlations for fracture characterization based on dielectric response, formation matrix properties, and fluid saturation.

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